

Introduction

In the field of digital animation in general and computer animation in particular, the phenomenon of rigging or skeletal animation refers to the technique wherein an entity, which is mostly a designated character or any other articulated and arbitrary object, is depicted in two portions. One among these portions is constituted by the surface representation that is made use of to draw and sketch out the character in the first place. In animation parlance, this bit is addressed as the skin or the mesh of the character in consideration. The second portion comprises a series of interconnected and interrelated parts that abide by a very specific hierarchical arrangement. This bit, in animation circles, is referred to as bones.

A set of bones comes together to collectively constitute and form the skeleton, which is otherwise popularly called as the rig of the character that has to be animated. The purpose of these bones is to play the role of a virtual armature that can be utilized to execute various animation manoeuvres upon the mesh. Generally, these manoeuvres take the form of defining and shaping the keyframe and pose of the skin; segments that are often regarded to be utterly pivotal concerning the framework of any mesh out there. In essence, digital rigs comprise the virtual anatomy including but not limited to bones, muscles, and joints, which in turn renders a model its locomotive abilities. Purely in terms of functionality, it draws an analogy with the strings that are used to control a marionette. Using Adobe Animate, which is Adobe Systems' exclusive computer animation and multimedia authoring program, one can easily execute the process of rigging in their animated projects.

Purpose of rigging

Though traditionally the process of rigging is more or less used to carry out the animation of sentient beings such as humans, and other related organic forms, it also serves as an excellent means of making the overall process of animation highly intuitive and extremely spontaneous. By making use of the technique of skeletal animation, one can also exercise complete control over carrying out the deformation of any arbitrary object out there. Irrespective of whether it is a spoon, a door, a building, or even a galaxy, if one is well acquainted with the nuances and inherent instructions that constitute rigging, they can just as easily cause the deformation of these objects.

An important point to be noted here is that when the animated object fails to fall under the category of an organic item in general, and a humanoid